

The Algebraic Bridge: FFGFT and $\mathbb{Z}_3\mathbb{C}$

Frobenius Automorphism, Charge Classification and
Five Structural Questions between $T^4/\mathbb{Z}_3$ and $\text{GF}(9)$

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Contents

The Algebraic Bridge: FFGFT and $\mathbb{Z}_3\mathbb{C}$	2
Notation	1
.1 Motivation: Characteristic 3 and FFGFT	2
.2 Five structural questions — updated status	2
Question 1: Number operator N and FFGFT winding quantum Δ_W	2
Question 2: Charge formula $Q = N/3$ and $\mathbb{Z}_3$ classification	2
Question 3: Sector pairing $k \leftrightarrow -k$ and Frobenius automorphism	3
Question 4: Casimir singularity and ξ in $\text{GF}(3)$	3
Question 5: $\sqrt{2}$ in $\text{GF}(9)$ and n_{thresh}	3
.3 Three further structural results	4
Frobenius fixed points = FFGFT sectors (Question 3, extended)	4
Norm hierarchy $\text{GF}(3) \rightarrow \text{GF}(9) \rightarrow \mathbb{R}(\xi)$ for masses	4
Weinberg angle: $\text{GF}(9)$ gives topology, ξ gives value	4
.4 Status overview	5
.5 Five algebraic computations (script 328)	5
Gluon representations from bilaterals	5
Tripotent classification	5
Generator comparison: FFGFT trine vs. Matzke's $(1 + 1j)^n$	5
$\text{GF}(27)$ and three generations	5
Noether conserved charge of the tripotent symmetry	6
.6 CKM Wolfenstein hierarchy from ξ and $\text{GF}(27)$ (script 329)	6
$\lambda = \xi^{1/6}$	6
$ V_{ub} = \sqrt{\xi} \cdot (C_2 - 1) = \sqrt{\xi}/3$	6
Consistency: $A \cdot \vec{p} \approx 1/3$	7
.7 ξ as a Galois Number	7
Core identity	7
Winding numbers from $\text{GF}(3^n)$	7
Verification scripts	8
AI note	8
References	8

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Abstract

Two independently developed frameworks — the Fundamental Fractal Geometric Field Theory (FFGFT) with its compact $T^4/\mathbb{Z}_3$ geometry and Matzke's GALG with ternary algebra $\mathbb{Z}_3\mathbb{C}$ — arrive at the same $\mathbb{Z}_3$ structure by entirely different routes. This document shows algebraically *why* this is not a coincidence, and derives new physical relations from it.

The key is the *Frobenius automorphism* $x \mapsto x^3$ in GF(9): it simultaneously realises the FFGFT sector pairing $k \leftrightarrow -k$, Matzke's vacuum conjugation $V_s = V^\dagger$, and the field conjugation in $\mathbb{Z}_3\mathbb{C}$ **[B]**.

The main new results:

- The Wolfenstein parameter of the CKM matrix satisfies $\lambda = \xi^{1/6}$ (0.5% deviation), where the exponent $1/6$ follows directly from the Galois tower structure $\text{GF}(3) \subset \text{GF}(9) \subset \text{GF}(27)$ **[K]**.
- The CKM element $|V_{ub}| = \sqrt{\xi}/3 = \sqrt{\xi} \cdot (C_2 - 1)$ (0.8% deviation), where $C_2 = 4/3$ is the Casimir value of SU(3) **[K]**.
- GF(9) and GF(27) classify the particle spectrum topologically — but exact mass values require $\xi \in \mathbb{R}$ **[B]**.

Notation: Symmetric $\mathbb{Z}_3\mathbb{C}$ after Matzke: $a + bi$ with $a, b \in \{-1, 0, +1\}$, $+1 + 1 = -1$ (ternary inline carry).

Guide: what this document shows

The basic problem. The three quark generations mix with strongly different strengths. The Wolfenstein parameter $\lambda \approx 0.225$ describes this hierarchy empirically — but why exactly this value? Why three generations? Why do the muon and electron have such different masses, even though both are leptons?

What GF(9) and GF(27) contribute. The finite field GF(9) has exactly 9 elements. Its Frobenius automorphism $x \mapsto x^3$ partitions them into three classes: fixed points $\{0, +1, -1\}$ (the FFGFT sectors), and three conjugate pairs $(k, -k)$ (the sector pairing). GF(27) extends this: its Galois tower over GF(3) has degree 6, and the 8 three-cycles correspond to 8 quark-colour states in 3 generations.

What ξ does that GF(9) cannot. GF(9) is a finite field — it has exactly 9 elements, and none of them equals $m_e/m_p \approx 4 \times 10^{-23}$. Exact particle masses cannot be represented in a finite field. The transition to physical scales occurs through $\xi = 4/30\,000$, whose denominator 3 forms a singularity in GF(3) — this is algebraically the transition from discrete topology to continuous physics.

The surprising part. Despite this, the combination of GF tower structure and ξ yields precise numerical predictions for CKM parameters that were previously known only empirically. This suggests that the Wolfenstein hierarchy is not a free parametrisation but follows from algebraic structure.

Notation and symbols

Symbol	Meaning	Source
<i>Status markers</i>		
[B]	algebraically proved, verification script available	Doc. 190
[K]	derivable from ξ (internally consistent)	Doc. 190
[S]	open bridge (sketch, no proof yet)	Doc. 190
<i>Fields and algebras</i>		
GF(p^n)	Finite field with p^n elements	Galois theory
$\mathbb{Z}_3\mathbb{C}$	Sym. GF(9): $a + bi$, $a, b \in \{-1, 0, +1\}$	[1, 3]
$\phi : x \mapsto x^3$	Frobenius automorphism in GF(9)	dieses Dok.
$\varphi : N \mapsto N \bmod 3$	Ring homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}$	dieses Dok.
<i>FFGFT quantities</i>		
ξ	$= 4/30\,000$, geometric base parameter	[9]
Δw	Winding quantum on $T^4/\mathbb{Z}_3$, $\Delta w \in \{0, +1, -1\}$	[10]
$T^4/\mathbb{Z}_3$	Compact 4-torus with $\mathbb{Z}_3$ orbifold	[9]
$k \leftrightarrow -k$	Sector pairing on $T^4/\mathbb{Z}_3$	[10]
<i>Matzke GALG</i>		
$V_s = V^\dagger$	Vacuum conjugation (adjunction)	[11]
$T_k^3 = \mathbf{1}$	Trine relation (tripotent condition)	[11]
$N = \sum q_k^\dagger q_k$	Number operator, $\sigma(N) = \{0, 1, 2, 3\}$	[4, 1]
<i>CKM notation</i>		
$\lambda \approx 0.225$	Wolfenstein parameter (Cabibbo angle)	[8]
$C_2 = 4/3$	Casimir value $SU(3)_{\text{fund}}$	Lie algebra
$ V_{ub} $	CKM element: $u \rightarrow b$ transition	[8]

Symmetric $\mathbb{Z}_3\mathbb{C}$: Elements $a + bi$ with $a, b \in \{-1, 0, +1\}$ use the symmetric representation of GF(3) around zero. $+1 + 1 = -1$ (ternary inline carry) and $-1 - 1 = +1$ (ternary inline borrow). The generator $(1 + i)^n$ has order 8.

.1 Motivation: Characteristic 3 and FFGFT

In characteristic 3, $(1 + N)^3 = 1$ for nilpotent N with $N^2 = 0$ holds *algebraically exactly* — because $\binom{3}{k} \equiv 0 \pmod{3}$ for $0 < k < 3$. The machine-epsilon result $|T_k^3 - \mathbf{1}| < 6.5 \times 10^{-16}$ from Doc. 324 is therefore not an approximation but a consequence of an algebraic theorem **[B]**.

.2 Five structural questions — updated status

Question 1: Number operator N and FFGFT winding quantum Δ_w

Previously open **[[S]]**; now partially closed.

Furey's number operator $N = \sum_j q_j^\dagger q_j$ has spectrum $\{0, 1, 1, 1, 2, 2, 2, 3\}$ on the eight basis states of one generation. Reduction mod 3 gives:

$$N \bmod 3 = \{0, +1, +1, +1, -1, -1, -1, 0\} \quad (1)$$

with particle classification:

$N \bmod 3 = 0$: colour-neutral (neutrino, positron)

$N \bmod 3 = 1$: anti-down quarks (3 colours)

$N \bmod 3 = 2$: up quarks (3 colours)

This is *exactly* the $\mathbb{Z}_3$ class decomposition of the FFGFT sectors $\Delta_w \in \{0, +1, -1\}$ on $T^4/\mathbb{Z}_3$.

Ring homomorphism [B]:

$$\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}, \quad \varphi(N) = N \bmod 3 = \Delta_w \quad (2)$$

is the canonical quotient; $\varphi(N + M) = \varphi(N) + \varphi(M) \bmod 3$ is a proved algebraic property.

Status: [B] (homomorphism), [K] (physical assignment)

$\varphi(N) = N \bmod 3$ is a ring homomorphism **[B]**. The physical identification $\varphi(N) = \Delta_w$ (Furey's number operator $\leftrightarrow$ FFGFT winding quantum) classifies the same threefold particle spectrum decomposition; the deeper algebraic reason for this agreement is **[K]**.

Question 2: Charge formula $Q = N/3$ and $\mathbb{Z}_3$ classification

Previously open **[[S]]**; now partially closed.

$Q = N/3 \in \mathbb{Q}$ cannot be embedded into GF(3) (3 is a zero divisor in $\mathbb{Z}/9\mathbb{Z}$, no inverse). However an injection exists:

$$\iota : \{0, \frac{1}{3}, \frac{2}{3}, 1\} \hookrightarrow \mathbb{Z}/3\mathbb{Z} \times \{0, 1\}, \quad \iota(Q) = (3Q \bmod 3, [Q]) \quad (3)$$

with values:

Q	$\iota(Q)$	Particle
0	(0, 0)	Neutrino
1/3	(1, 0)	Anti-Down
2/3	(2, 0)	Up
1	(0, 1)	Positron

The $\mathbb{Z}/3\mathbb{Z}$ component of $\iota(Q)$ is exactly $N \bmod 3$.

Status: [B] (classification), [K] (Q as $\mathbb{Q}$ number)

$N \bmod 3$ fully classifies $Q = N/3$ **[B]**. Q itself (as rational number $1/3, 2/3$) requires $\mathbb{Q}$ and is not a GF(3) object **[K]**.

Question 3: Sector pairing $k \leftrightarrow -k$ and Frobenius automorphism

Main result; was [S]; now [B].

The Frobenius automorphism in GF(9):

$$\phi : x \mapsto x^3, \quad \phi(a + bi) = a - bi \quad [\text{field conjugation}] \quad (4)$$

operates simultaneously on three levels:

- **GF(9) field:** $a + bi \mapsto a - bi$ (conjugation)
- **FFGFT sector:** $k \mapsto -k \bmod 3$ ($\mathbb{Z}_3$ inversion)
- **Matzke vacuum:** $V \mapsto V_s = V^\dagger$ (matrix adjunction)
- `pruef_326_gf9_orbit_masse_weinberg.py`: Fixed points **[B]**, norm hierarchy **[B]**, Weinberg topology **[K]**
- `pruef_328_gluonen_tripotente_gf27_noether.py`: gluons **[B]**, tripotents **[B]**, GF(27) **[K]**, Noether **[B]**
- `pruef_329_ckm_xi_gf27.py`: $\lambda = \xi^{1/6}$, $|V_{ub}| = \sqrt{\xi}/3$ **[K]**

Numerically verified: $x^3 = \bar{x}$ for all 8 non-zero GF(9) elements **[B]**.

Status [B]

All three operations are algebraically isomorphic realisations of the Frobenius automorphism $x \mapsto x^3$.

Question 4: Casimir singularity and ξ in GF(3)

$\xi = (4/3)/10^4$: Casimir denominator $3 \equiv 0$ in GF(3) is singular. $\xi \bmod 3 = 1$ — ξ lies in residue class 1. The singularity marks the transition $\text{GF}(3) \rightarrow \mathbb{R}$ **[K]**.

Question 5: $\sqrt{2}$ in GF(9) and n_{thresh}

$i^2 = -1 \equiv 2 \pmod{3}$, so $\sqrt{2} = i \in \text{GF}(9)$ **[B]**. $n_{\text{thresh}} = 2\pi/(\sqrt{2} \ln 2) \approx 6.41$ (R93) thus has an algebraically exact GF(9) kernel.

.3 Three further structural results

Frobenius fixed points = FFGFT sectors (Question 3, extended)

The complete orbit structure under $\phi(x) = x^3$ in $\text{GF}(9)$:

- **Fixed points** ($x^3 = x$): $\{0, +1, -1\} = \text{GF}(3)$ — exactly the FFGFT sectors $\Delta_w \in \{0, +1, -1\}$ **[B]**
- **Non-fixed points**: 6 elements in 3 conjugate 2-cycles **[B]**: $(i \leftrightarrow 2i), (1+i \leftrightarrow 1+2i), (2+i \leftrightarrow 2+2i)$ — exactly the FFGFT pairs $(k, -k)$ for $k \neq 0$.

Stable states under Frobenius are the sectors; unstable states are the conjugate pairs.

Norm hierarchy $\text{GF}(3) \rightarrow \text{GF}(9) \rightarrow \mathbb{R}(\xi)$ for masses

In $\text{GF}(3)$, $1^2 \equiv 2^2 \equiv 1 \pmod{3}$ — muon and tau are *algebraically degenerate* as squares **[B]**. The $\text{GF}(3)$ structure distinguishes only:

$$n^2 \in \{0, 1\} \pmod{3} : \quad 0 \leftrightarrow \text{massless}, \quad 1 \leftrightarrow \text{massive (generations 2+3 degenerate)} \quad (5)$$

In $\text{GF}(9)$, the norm $\|x\| = (a^2 + b^2) \pmod{3}$ gives two classes **[B]**:

$$\begin{aligned} \|x\| = 1 & : \{+i, -i, +1, -1\} \quad \text{— purely real/imaginary (light)} \\ \|x\| = -1 & : \{+1+i, +1-i, -1+i, -1-i\} \quad \text{— mixed (heavy)} \end{aligned}$$

The full mass hierarchy requires ξ **[K]**: ξ breaks the $\text{GF}(3)$ degeneracy and gives $m_\mu/m_e \approx 207$, $m_\tau/m_\mu \approx 17$ as continuous numbers.

Status: **[B]** (degeneracy), **[K]** (mass values)

Hierarchy: $\text{GF}(3)$ gives whether massive/massless; $\text{GF}(9)$ gives light/heavy; $\xi \in \mathbb{R}$ gives exact mass ratios. The singularity $C_2 = 4/3$ in $\text{GF}(3)$ (Question 4) marks exactly the transition $\text{GF}(3) \rightarrow \mathbb{R}$.

Weinberg angle: $\text{GF}(9)$ gives topology, ξ gives value

$\text{Gal}(\text{GF}(9)/\text{GF}(3)) = \mathbb{Z}_2$ explains *why* there are two gauge groups $\text{SU}(2)_L$ and $\text{U}(1)_Y$ — a degree-2 extension gives exactly two components **[K]**.

The best $\text{GF}(9)$ candidate for $\sin^2 \theta_W$ is $|\text{GF}(3)^*|/|\text{GF}(9)^*| = 2/8 = 0.250$, compared to the experimental value 0.231 (8% deviation). The exact value is reached by ξ corrections (Doc. 323, **[K]**).

Addendum (Doc. 372, 21 Sept. 2026). The candidate $1/4$ is superseded by the icosahedral matrix element $|\langle v_0 | R_5 | v_1 \rangle|^2 = 2/9$: the on-shell Weinberg angle $1 - M_W^2/M_Z^2$ lies within 0.4% of $2/9$ (Doc. 372 §3.1). The comparison value 0.231 above is the $\overline{\text{MS}}$ value; for pole masses from $\tilde{T} \cdot m = 1$ the on-shell scheme is the natural one. The scale flow from ξ (Doc. 323) describes the $\overline{\text{MS}}$ scheme and stands. The $\text{GF}(9)$ statement of this section — a degree-2 extension explains the two gauge groups — is unaffected.

Status **[K]**

$\text{GF}(9)$ provides the algebraic justification for two gauge groups (degree-2 Galois extension). The numerical value follows in the on-shell scheme from the A_5 matrix element $2/9$ (Doc. 372), in the $\overline{\text{MS}}$ scheme from the ξ scale flow (Doc. 323).

.4 Status overview

Table 1: Algebraic bridge questions FFGFT $\leftrightarrow$ GF(9) — updated status

#	Question	Result	Status
1a	Ring homo $\varphi(N) = N \bmod 3$	canonical, proved	[B]
1b	$\varphi(N) = \Delta_w$ physical	same threefold split	[K]
2a	$N \bmod 3$ classifies Q	injection ι proved	[B]
2b	$Q = N/3$ as GF(3) object	impossible ($3 \equiv 0$)	[K]
3	$k \leftrightarrow -k$ vs. $V_s = V^\dagger$	Frobenius $x \mapsto x^3$	[B]
4	$C_2 = 4/3$ and ξ in GF(3)	singularity = GF(3) $\rightarrow \mathbb{R}$	[K]
5	$\sqrt{2}$ in GF(9), n_{thresh}	$i^2 = 2$, kernel exact	[B]
P3b	Fixed pts. = sectors, $(k, -k)$ pairs	orbit structure	[B]
P4b	$1^2 = 2^2 = 1$ in GF(3)	Gen. 2+3 degenerate	[B]
P4c	Norm $\in \{1, 2\}$ in GF(9)	light/heavy	[B]
P5b	$ \text{GF}(3)^* / \text{GF}(9)^* = 1/4$	tree-level $\sin^2 \theta_W$	[K]

.5 Five algebraic computations (script 328)

Gluon representations from bilaterals

With the normalised number operator $N = \sum_k q_k^\dagger q_k$ (Furey normalisation $q_k = a_{p,k}/2$), the spectrum is $\sigma(N) = \{0, 1, 2, 3\}$ **[B]** — exactly Furey's result.

The *bilaterals* $q_i q_j^\dagger$ yield gluon representations:

- 6 off-diagonal $(q_i q_j^\dagger, i \neq j)$: traceless, rank 2 **[B]**
- 2 diagonal from $(N_i - N_j)/\sqrt{2}$: SU(3) Cartan generators **[B]**

The 8 matrices are linearly independent (rank 8) **[B]** — 8 gluon representations matching the 8 generators of SU(3).

Tripotent classification

All tripotents T with $T^3 = T$ in the 8-dimensional representation:

- $T_k = P_{p,k} - P_{n,k}$: eigenvalues $\{+1, -1\}$ in sym. $\mathbb{Z}_3\mathbb{C}$ **[B]**
- $P_{p,k}, P_{n,k}$ themselves: tripotents with eigenvalues $\{0, +1\}$ **[B]**
- 6 tripotent products $T_i \cdot T_j$ ($i \neq j$) **[B]**

Generator comparison: FFGFT trine vs. Matzke's $(1 + 1j)^n$

FFGFT trine: $T_k^3 = \mathbf{1}$ to 6.5×10^{-16} **[B]** (algebraically exact in char. 3, Doc. 324). Matzke's $(1 + 1j)$ analogue in 8×8 : $D^8 = 256 \cdot \mathbf{1}$ (scaling factor 256 from the matrix representation). In sym. $\mathbb{Z}_3\mathbb{C}$: $(1 + i)^8 = 1$ (order 8) **[B]**.

GF(27) and three generations

GF(27) = GF(3)[x]/($x^3 + 2x + 1$) (irreducible, also $x^3 + 2x + 2$) **[B]**.

Orbit structure of Frobenius $x \mapsto x^3$ in $\text{GF}(27)$: 3 fixed points = $\text{GF}(3)$, $(27 - 3)/3 = 8$ three-cycles **[B]**.

Status [K]

8 three-cycles in $\text{GF}(27)$ correspond to 8 quark-colour states in 3 generations. The Galois group $\text{Gal}(\text{GF}(27)/\text{GF}(3)) = \mathbb{Z}_3$ may describe generation mixing (CKM matrix) **[S]**.

Noether conserved charge of the tripotent symmetry

The tripotent condition $X^3 = X$ is a $\mathbb{Z}_3$ -affine symmetry. By Noether's theorem, every symmetry generates a conserved charge.

The corresponding Noether charge is **[B]**:

$$Q_{\mathbb{Z}_3} = N \bmod 3 = \Delta_w \quad (6)$$

This is identical to the ring homomorphism φ from §2.1. The tripotent condition $X^3 = X$ is the algebraic form of the discrete $\mathbb{Z}_3$ symmetry of the orbifold $T^4/\mathbb{Z}_3$.

Status [B]

Tripotent symmetry ($X^3 = X$) $\Leftrightarrow$ FFGFT $\mathbb{Z}_3$ orbifold symmetry. Noether charge = FFGFT winding quantum Δ_w = colour charge class.

.6 CKM Wolfenstein hierarchy from ξ and $\text{GF}(27)$ (script 329)

From the $\text{GF}(27)$ Galois tower structure and ξ , three precise relations emerge for the CKM Wolfenstein parameters.

$$\lambda = \xi^{1/6}$$

The Galois tower $\text{GF}(3) \subset \text{GF}(9) \subset \text{GF}(27)$ has degrees $[\text{GF}(9) : \text{GF}(3)] = 2$ and $[\text{GF}(27) : \text{GF}(3)] = 3$, total degree $2 \cdot 3 = 6$. This gives **[K]**:

$$\lambda = \xi^{1/6} = \xi^{1/(2 \cdot 3)} = 0.2260 \quad (\text{exp.: } \lambda = 0.2250, \text{ dev. } 0.5 \%) \quad (7)$$

The Wolfenstein hierarchy is the ξ hierarchy in steps of $1/6$: $|V_{us}| \sim \xi^{1/6}$, $|V_{cb}| \sim \xi^{1/3}$, $|V_{ub}| \sim \xi^{1/2}$.

$$|V_{ub}| = \sqrt{\xi} \cdot (C_2 - 1) = \sqrt{\xi}/3$$

With $C_2 = 4/3$ (Casimir value $\text{SU}(3)_{\text{fund}}$), $C_2 - 1 = 1/3$ **[B]**, and:

$$|V_{ub}| = \sqrt{\xi} \cdot (C_2 - 1) = \frac{\sqrt{\xi}}{3} = 3.85 \times 10^{-3} \quad (\text{PDG: } 3.82 \times 10^{-3}, \text{ dev. } 0.8 \%) \quad \textbf{[K]} \quad (8)$$

This relation connects the $\text{GF}(9)$ extension ($\sqrt{\xi}$) with the $\text{SU}(3)$ Casimir value.

Consistency: $A \cdot |\vec{\rho}| \approx 1/3$

From $\lambda = \xi^{1/6}$ and $|V_{ub}| = \sqrt{\xi}/3$ it follows necessarily **[K]**:

$$A\sqrt{\rho^2 + \eta^2} = \frac{|V_{ub}|}{\lambda^3} = \frac{\sqrt{\xi}/3}{\xi^{1/2}} = \frac{1}{3} = C_2 - 1 \quad (9)$$

Numerically: $A\sqrt{\rho^2 + \eta^2} = 0.316$ vs. $1/3 = 0.333$ (5% dev.).

Status [K]

CKM Wolfenstein hierarchy from GF tower structure and ξ :

$$\lambda = \xi^{1/(\deg(\text{GF}^9) \cdot \deg(\text{GF}^{27}))} = \xi^{1/6}$$

$$|V_{ub}| = \sqrt{\xi} \cdot (C_2 - 1) = \sqrt{\xi}/3$$

Verification script: `pruef_329_ckm_xi_gf27.py` (16 assertions, all OK).

.7 ξ as a Galois Number

The mass-ratio identity from §?? leads to a new result: $\xi = 4/30000$ is not a free constant but a Galois number **[K]**.

Core identity

The FFGFT formula (Doc. 006) and the Galois group orders from §?? yield:

$$\frac{(r_\mu/r_e)^2}{\xi} = |\text{GF}(9)^*|^2 \cdot 5^2 \cdot |\text{GF}(27)| = 8^2 \cdot 25 \cdot 27 = 43200 \quad (10)$$

These are not two computations of the same value but a single algebraic equation. From it ξ follows:

$$\xi = \frac{(r_\mu/r_e)^2}{43200} = \frac{(12/5)^2}{43200} = \frac{1}{7500} = \frac{4}{30000} \quad [\text{K}] \quad (11)$$

Winding numbers from $\text{GF}(3^n)$

The quantum numbers $r_g = n_{\phi,g}^2/n_{\theta,g}$ (Doc. 317) follow from a recursion with a Galois initial condition:

Winding number	Value	Galois origin
$n_{\theta,1}$	3	$\text{char}(\text{GF}(3^n))$
$n_{\phi,1}$	2	$ \text{GF}(3)^* $
$n_{\theta,2}$	5	recursion = 3 + 2; min. prime factor of $ \text{GF}(81)^* $
$n_{\phi,2}$	4	
$n_{\theta,3}$	9	recursion = 5 + 4
$n_{\phi,3}$	5	min. prime factor of $ \text{GF}(81)^* = 80$

All factors in the core identity are Galois orders: $8 = |\text{GF}(9)^*|$, $27 = |\text{GF}(27)|$, $5 = \text{min.prime}(|\text{GF}(81)^*|)$. Verification scripts: `pruef_330` (mass ratio), `pruef_331` (ξ derivation).

Verification scripts

- `pruef_324_z3c_trine_exakt.py`: Algebraic proof $T_k^3 = 1$ in char. 3 **[B]**
- `pruef_325_gf9_ffgft_bruecke.py`: All five questions; Frobenius **[B]**; ring homo **[B]**; injection ι **[B]**

Note on AI assistance

This document was produced with AI-assisted support (Claude, Anthropic). All algebraic results were independently verified by Python verification scripts; the author bears sole responsibility for the content.

The methodological principles governing the use of AI in FFGFT documents are described in the A-series (in particular Docs. A001–A010): AI-generated content is only incorporated into the corpus after independent review and verification by scripts. External criticism and AI outputs are not adopted without scrutiny (Doc. 190, methodology note).

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